

Enhancing continuous-flow CO₂ capture and release from aqueous carbonates via thermal pH regulation

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Current carbon capture technologies face a critical tradeoff between CO₂ absorption capacity and regeneration energy requirements. Here we present an approach to reduce the regeneration energy load by utilizing tris(hydroxymethyl)aminomethane (Tris) as a thermally responsive pH regulator in aqueous carbonate solutions. By leveraging the temperature-dependent equilibrium constant of Tris, our system achieves enhanced CO₂ absorption under ambient conditions and enables low-temperature CO₂ desorption (≤ 60 °C, 1 atm) through controlled pH swings. A continuous-flow reactor demonstrates the efficient concentration of diluted CO₂ streams (1–5%) to high-purity products with low energy inputs (even solely with natural sunlight), high stability (>240 h) and promising economic viability. This work represents a sustainable carbon capture technology that offers a practical pathway for industrial-scale implementation while minimizing energy penalties associated with traditional CO₂ capture processes.

Anthropogenic carbon dioxide (CO₂) emissions have reached unprecedented levels, surpassing 36 gigatons annually in recent years¹. This relentless accumulation of greenhouse gases in the atmosphere has driven an alarming acceleration in global climate change, with average temperatures now exceeding 1.2 °C above pre-industrial levels². Such rapid warming has triggered a series of devastating environmental impacts, from intensifying extreme weather and prolonged droughts to accelerating sea-level rise and biodiversity loss. The escalating climate crisis has spurred global initiatives to develop carbon mitigation strategies, with carbon capture and storage technologies becoming a cornerstone of decarbonization efforts. The Intergovernmental Panel on Climate Change has identified large-scale carbon capture and storage deployment as essential for achieving net-zero emissions by 2050—both through permanent geological sequestration and the conversion of captured CO₂ into value-added products—to meet the critical 1.5 °C warming threshold^{3–5}.

CO₂, as an acidic gas, readily undergoes acid–base neutralization reactions with basic solutions, forming dissolved carbonate and

bicarbonate species. This fundamental chemical principle has been leveraged in pH-swing capture strategies, where alkaline absorbents such as metal hydroxides and carbonates serve as efficient CO₂ capture media. Compared with conventional aqueous or solid amines, basic carbonate solutions offer distinct advantages, including low corrosion, superior thermal and oxidative resistance and environmental benignity. These properties demonstrate their versatility in capturing CO₂ from both concentrated point sources ($\sim 15\%$) to ultradilute atmospheric conditions ($\sim 0.04\text{--}1\%$)^{6–9}. However, traditional thermal regeneration typically requires temperatures exceeding 140 °C at 1 atm pressure for carbonate systems, resulting in substantial energy penalties that hinder their economic viability for large-scale deployment^{7,10,11}.

Extensive research efforts have been devoted to enhancing the pH-swing processes for effective carbon capture from diverse sources such as industrial facilities¹², oceanwater^{13–15} and the atmosphere^{16,17}. For CO₂ sources with relatively high concentrations, such as emissions from steel production and power plants, the incorporation of amine promoters has successfully lowered the desorption temperature by $\sim 10\text{--}20$ °C

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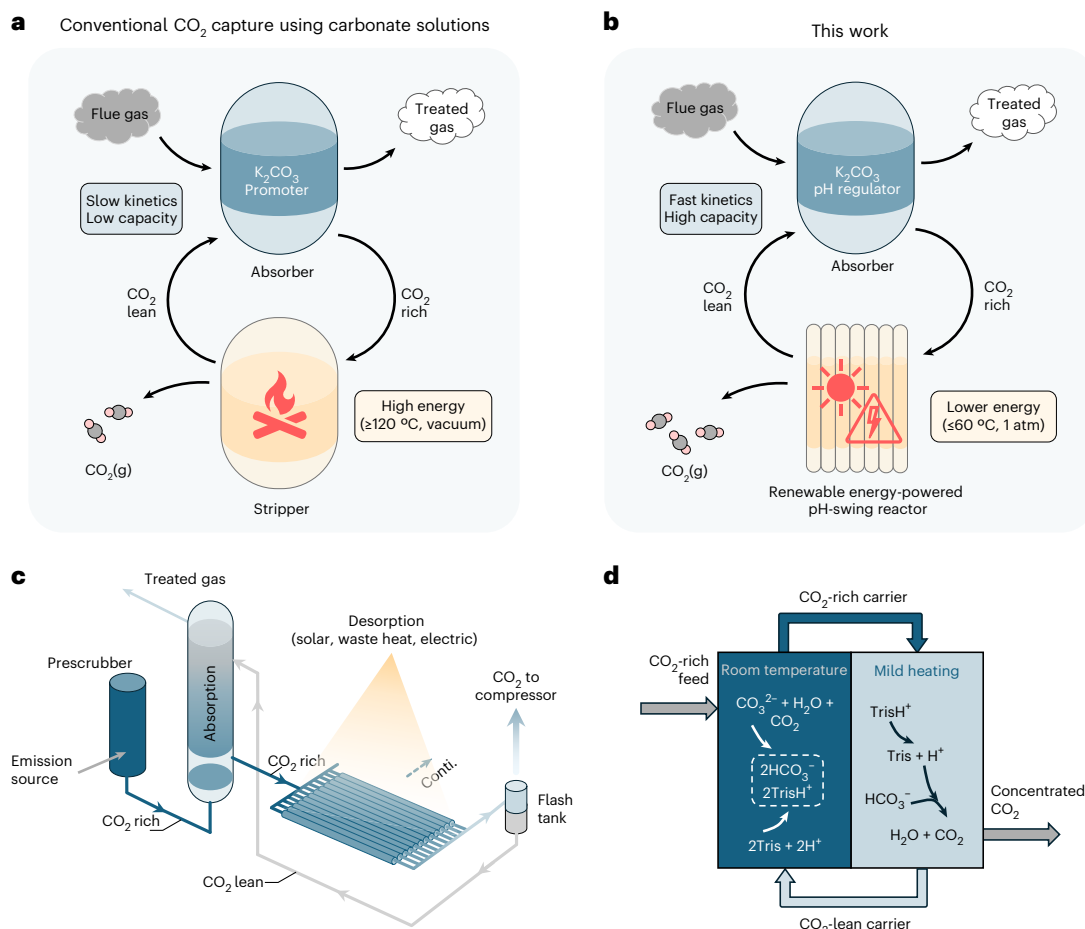


Fig. 1 | Overview of a thermal pH-swing carbon capture system with lowered energy inputs. **a**, An industrialized hot potassium carbonate carbon capture process with an energy-intensive regeneration process, as potassium bicarbonate solution requires heating exceeding $-120\text{ }^{\circ}\text{C}$ or a vacuum to release captured CO₂. **b**, The thermal pH-swing CO₂ capture and release with potassium carbonate sorbent solution enabled by a pH regulator to boost the absorption capacity and lower the regeneration temperature to $-60\text{ }^{\circ}\text{C}$ under ambient pressure. **c**, An illustration of a thermal pH-swing CO₂ capture system

powered by solar energy, electricity and/or waste heat. Conti. indicates a continuation of the reactor array. **d**, The mechanisms of thermal pH-swing CO₂ capture with the thermally responsive pH regulator, Tris. In a CO₂ capture and release cycle, Tris stabilizes the pH of the solution during CO₂ capture to boost the absorption capacity and facilitates the release of captured CO₂ via acid dissociation to react with bicarbonate species under mild heating conditions.

with simultaneously accelerated absorption kinetics^{18–23}. Innovations in reactor design have also enhanced the process efficiency through improved gas–liquid contact and heat integration^{24,25}. Nevertheless, conventional thermal regeneration methods with high energy demands still dominate industrial applications (Fig. 1a and Supplementary Table 1), often accounting for more than 70% of total operational costs in commercial systems^{12,23}. More recently, electrochemical methods have emerged as promising alternatives for capturing CO₂ from low-concentration sources, operating through electrolysis-driven acidification of (bi)carbonate solutions under ambient conditions with reduced energy demands^{17,26–29}. Yet, many demonstrations have typically encountered limitations due to the complex separation of acidic and alkaline solutions^{30,31} or relatively low CO₂ regeneration rates (for example, small operational current densities)^{32,33}. Parallel investigations have explored photochemical approaches, utilizing photoacids to modulate the pH of the capture solution through controlled light wavelength and intensity^{34–36}. Although conceptually promising, many aspects of photoacids, including their solubility, stability and scalable synthetic routes, require further examination to improve overall carbon capture performance^{37,38}. There remains a critical need to develop innovative approaches that can dramatically reduce the energy requirements for the regeneration of capture solutions with

high CO₂ capture efficiency, which could fundamentally transform carbon capture technologies and accelerate their implementation across various industrial sectors.

Here, we address existing limitations in aqueous carbonate-based CO₂ capture technologies with a thermal pH regulation strategy, simultaneously enhancing CO₂ absorption capacity and lowering the regeneration energy (Fig. 1b,c). Tris(hydroxymethyl)aminomethane (Tris), a water-soluble buffer with the formula (HOCH₂)₃CNH₂, is used for thermal pH regulation owing to its strong temperature-dependent pK_a . During the CO₂ absorption process, the conjugated acid of Tris (TrisH⁺) acts as a buffer to balance the charge of bicarbonate anions and stabilize the pH of the solution, resulting in enhanced CO₂ absorption capacity at room temperature (Fig. 1d). At a slightly elevated temperature ($\leq 60\text{ }^{\circ}\text{C}$), the dissociation of TrisH⁺ lowers the pH of the aqueous system, effectively releasing CO₂ from bicarbonate species at ambient pressure. This results in a low regeneration energy of -125 kJ per mole of CO₂ captured from a 15% source. This mild heating can be readily realized with electricity, waste heat and/or natural sunlight. As a proof-of-concept, a continuous-flow reactor with a tube-in-tube configuration was demonstrated to produce concentrated and high-purity CO₂ products, depending on the sweep stream used, from low-concentration point sources (down to 1%) by harnessing natural sunlight ($\leq 1\text{ kW m}^{-2}$).

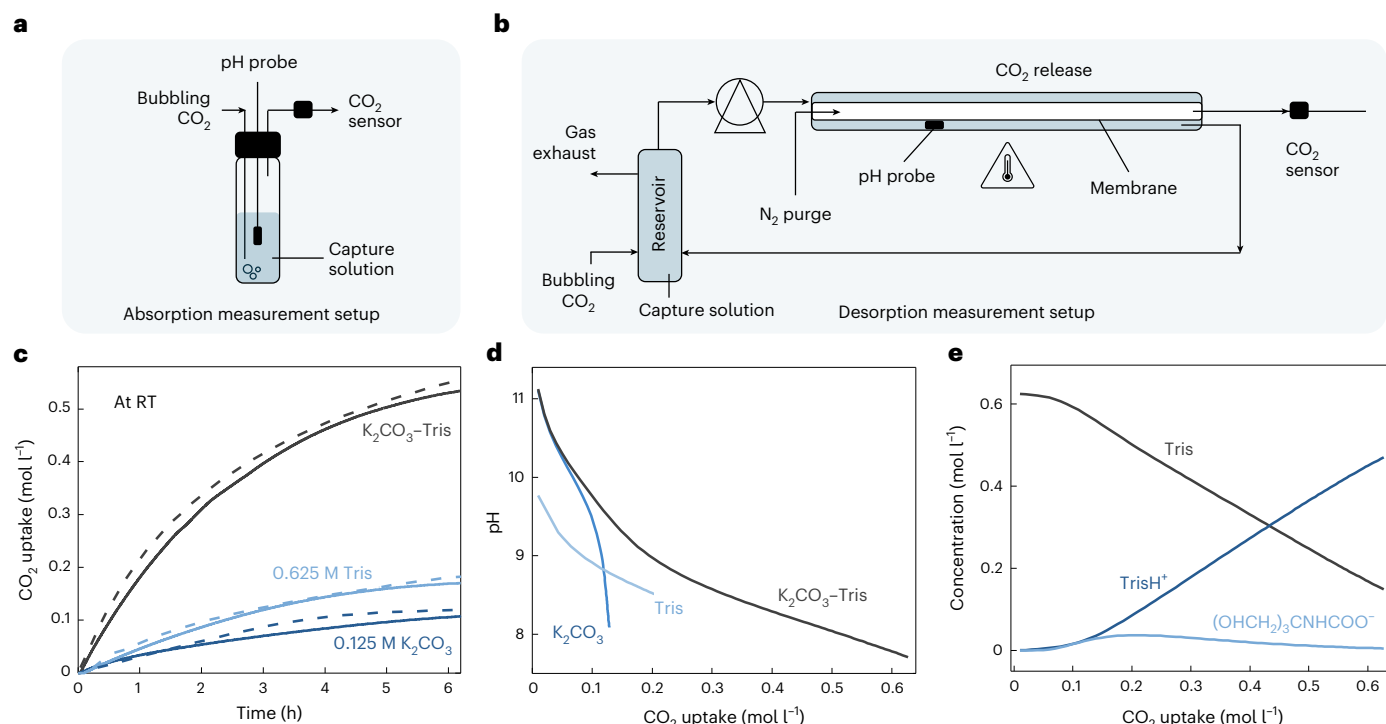


Fig. 2 | CO₂ absorption performance of the K₂CO₃-Tris carbon capture system.

a,b, Schematics of the absorption (**a**) and desorption (**b**) measurement setups.

c, The dynamic CO₂ absorption processes of different capture solutions measured at room temperature (RT) (solid lines) and calculated CO₂ uptake

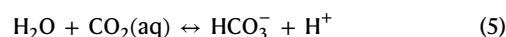
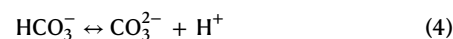
for each component (dashed lines). **d,** The calculated pH changes over CO₂ absorption for each component. **e,** The calculated concentrations of Tris species in the K₂CO₃-Tris aqueous system as a function of CO₂ uptake. RT, room temperature.

With high carbon capture rates, excellent stability (>10 days) and economic advantages, this work presents a seamless integration potential with existing industrial processes to achieve decarbonization goals using renewable energy sources.

Results

Thermal pH-swing system design and thermodynamic analysis

Bench-scale units were set up to investigate CO₂ capture and release performance in the K₂CO₃-Tris aqueous system (Methods and Fig. 2a,b), where pH probes and thermocouples were installed to monitor the in situ pH and temperature changes during the reactions. For a proof-of-concept demonstration, a relatively low-concentration capture solution (0.125 M K₂CO₃ and 0.625 M Tris) was selected on the basis of the largest pH changes associated with the temperature increase (Supplementary Table 2 and Supplementary Fig. 1). It should be noted that the pH of a CO₂-saturated sorbent solution should be lower than the pK_a of the selected acid at room temperature to maintain its protonated form but higher than the pK_a at elevated temperatures so that the acid becomes deprotonated, enabling a reversible pH-swing for CO₂ capture and release (Supplementary Fig. 2). The elementary steps involved in CO₂ absorption-desorption in the K₂CO₃-Tris aqueous system are presented in equations (1)–(6): the carbonate ions (CO₃²⁻) react with dissolved CO₂ and water (H₂O) to form bicarbonate ions (HCO₃⁻) (equations (4) and (5)), whereas Tris, (HOCH₂)₃CNH₂, acts as a buffer by accepting protons (H⁺) from water simultaneously (equation (1)). Tris, as a primary amine, also absorbs CO₂ to form a carbamate^{39,40} (equation (2)), with a much lower molar CO₂ capture capacity in comparison with K₂CO₃ (Supplementary Fig. 3). Under elevated temperatures, equation (1) plays a key role in assisting the pH swing to produce more protons (H⁺) in the solution, reacting with bicarbonate to release CO₂ (equations (3)–(6))



The dynamic absorption of CO₂ from a 5% CO₂ inlet stream bubbled into the K₂CO₃-Tris solution and control samples was measured at ambient pressure and -25 °C (Fig. 2c). The CO₂ absorption of the K₂CO₃-Tris system reached $-0.53 \pm 0.03 \text{ mol l}^{-1}$, which was much higher than that achieved by either of the single components or of their sum. This enhanced CO₂ capture capacity was ascribed to the buffer effect of Tris as it accepts protons (H⁺) to form TrisH⁺ to stabilize pH and balance the anionic HCO₃⁻, thereby facilitating a quasi-stable basic solution to absorb more CO₂ gas. Without Tris, the pH of the K₂CO₃ solution dropped to neutral after absorbing a small amount of CO₂ and was unable to capture more CO₂, leading to a lower absorption at $-0.12 \pm 0.02 \text{ mol l}^{-1}$. Speciation calculations during the CO₂ absorption process were carried out on the basis of the temperature-dependent equilibrium constants of each reaction, mass and charge balances and Henry's law⁴¹ (Supplementary Note 1). The calculated CO₂ capture capacities for all compositions are well matched with experimental results (Fig. 2c, dashed and solid lines, respectively). In addition, the calculated pH changes show a much slower decrease in the K₂CO₃-Tris aqueous system during CO₂ absorption compared with the sharp drop in pH with K₂CO₃ only (Fig. 2d), which confirms the buffering effect of

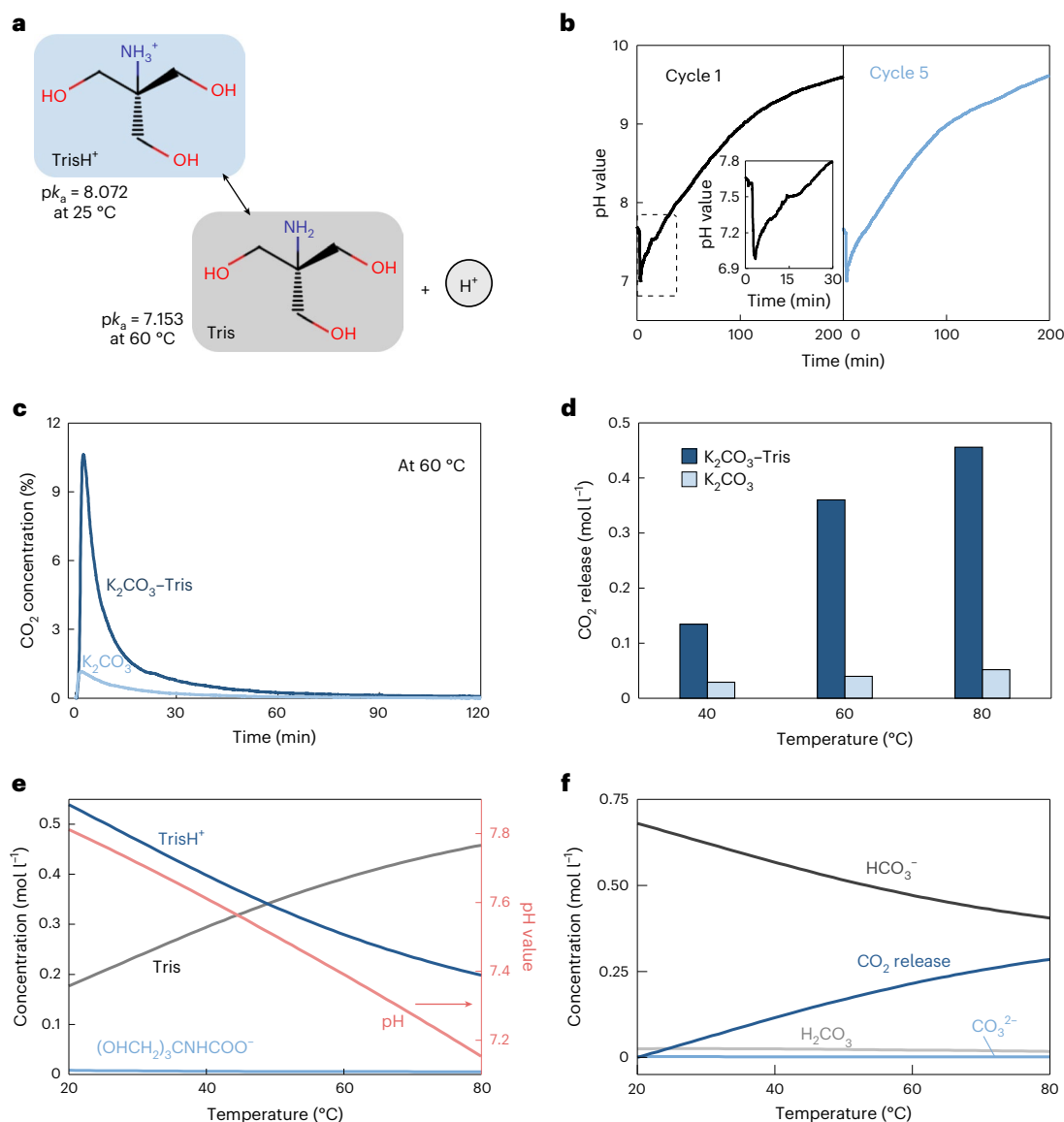


Fig. 3 | CO₂ release by Tris-enhanced thermal pH swing. **a**, A schematic of the temperature-dependent dissociation of TrisH⁺ in water. **b**, The dynamic pH changes of the CO₂-saturated K₂CO₃-Tris solution under mild heating at -60 °C for five cycles. The inset shows a zoom-in view during the heating stage. **c**, The desorption performance of the K₂CO₃-Tris system and K₂CO₃ only at 60 °C under ambient pressure. **d**, A comparison of the K₂CO₃-Tris system and K₂CO₃ only on

their CO₂ release performance at temperatures of 40 °C, 60 °C and 80 °C. **e, f**, The speciation calculations: the calculated pH change of the system and Tris species concentrations over 20–80 °C (**e**) and the calculated concentrations of carbon species over 20–80 °C (**f**). In **b–f**, the concentrations of K₂CO₃ and Tris were 0.125 M and 0.625 M, respectively.

Tris to boost the CO₂ absorption capacity. The buffer-enhanced CO₂ capture is further evidenced by the evolution of Tris species (Fig. 2e), where the TrisH⁺ concentration increases for pH buffering and charge balance during CO₂ absorption (Fig. 2e). Very little carbamate is formed, ruling out this pathway as a primary mechanism for CO₂ capture (Supplementary Fig. 4). The intrinsically high solubility of Tris and K₂CO₃ enables the use of high-concentration capture solutions (for example, 0.625 M of K₂CO₃ and 2 M of Tris) to achieve an even higher CO₂ absorption at -1.52 ± 0.04 mol l⁻¹ (Supplementary Figs. 5 and 6). As indicated by the overall CO₂ absorption reaction (Fig. 1d), a Tris/K₂CO₃ molar ratio of greater than 2 is sufficient to maintain effective buffering in the K₂CO₃-Tris system (Supplementary Fig. 7). Under batch-testing conditions, where CO₂ was bubbled through a solution in a vial, a greater than 70% CO₂ removal was achieved over the initial 3 h (Supplementary Fig. 8), a degree of removal that can be further improved with optimized contactors, such as countercurrent

flow packed columns. This experimental system was used solely for comparison of absorption capacities of different solutions, with no attempt made to optimize absorption kinetics.

Another key advantage of our system is the temperature-dependent pK_a of Tris, which enables the effective release of the captured CO₂ with reduced energy consumption. At elevated temperatures, the CO₂-saturated K₂CO₃-Tris solution exhibits a rapid and pronounced pH drop within 2 min due to the dissociation of TrisH⁺, followed by the pH increase along with the CO₂ release (Fig. 3a,b). This thermal pH swing shows excellent cyclability even after weeks of heating, indicating the high stability and durability of the sorbent solution (Supplementary Fig. 9). The pH of the K₂CO₃-Tris solution does not return to its original state before CO₂ absorption (Fig. 2d) owing to the residual CO₂ remaining in the solution, which enters subsequent cycles to reach a quasi-steady state and determine the working capacity of the system. The enhanced pH swing resulted in a higher amount of CO₂ release from

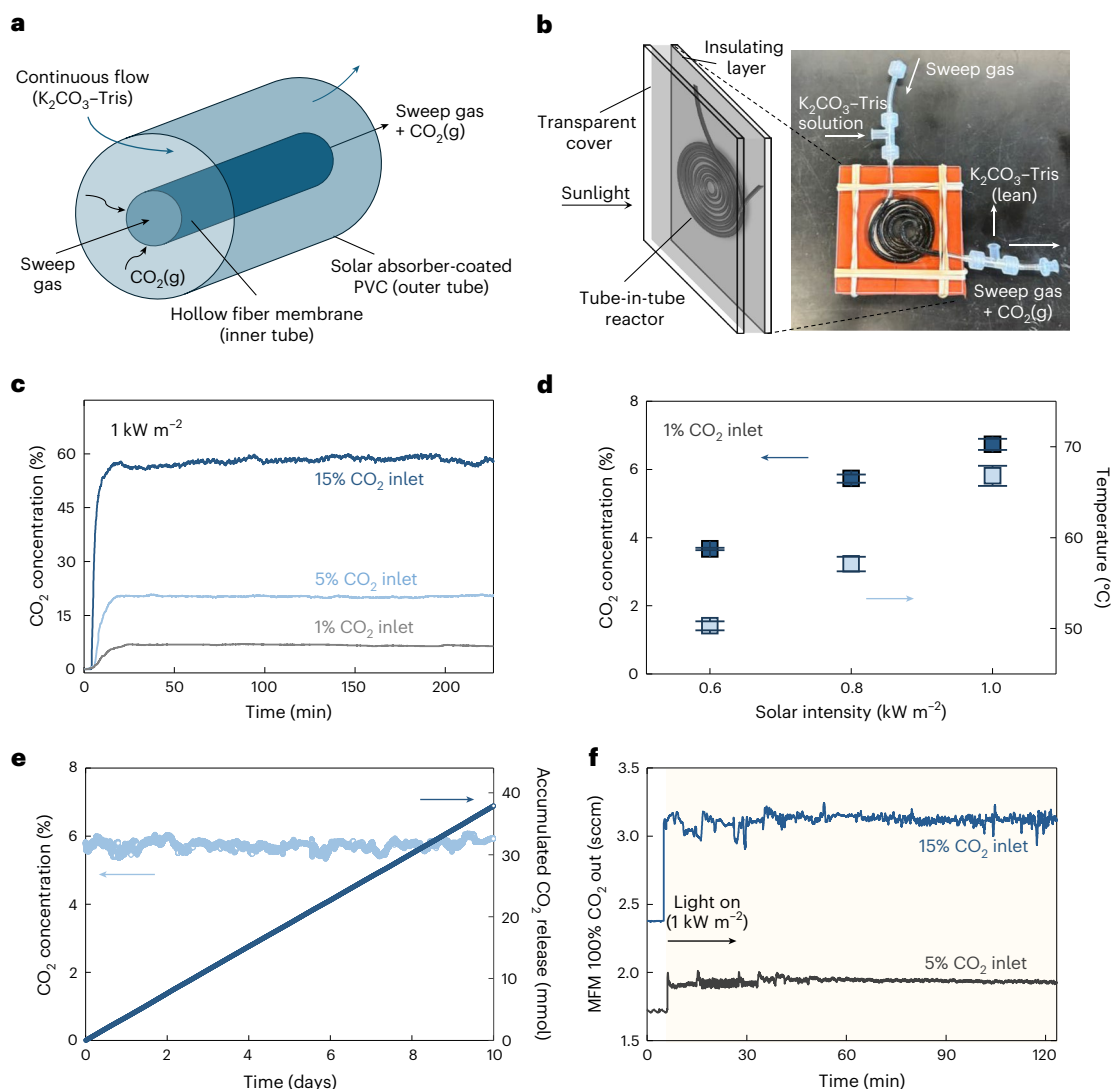


Fig. 4 | Solar-powered pH-swing CO_2 release. **a**, A schematic of tube-in-tube configuration for separating pure CO_2 from capture solution, K_2CO_3 -Tris. **b**, A schematic of designed solar-powered continuous-flow CO_2 release system. **c**, The CO_2 concentration performance from 1%, 5% and 15% CO_2 inlet sources using N_2 sweep gas under 1 sun (1 kW m^{-2}). **d**, The CO_2 concentration performance from 1% CO_2 inlet source under different solar intensities using N_2 sweep gas. The data are presented as mean $\pm$ s.d.; $n = 3$ measurements. **e**, The 10-day durability

of the solar-powered continuous-flow CO_2 capture and release system under 0.8 sun and the accumulated released CO_2 . Arrows in **(d)** and **(e)** denote the corresponding axes for each dataset in the same color. **f**, The MFM records of ~100% CO_2 released using the solar-powered continuous-flow system from solutions exposed to 5% and 15% CO_2 feeds to the absorber using pure CO_2 sweep gas. The arrow indicates the time point at which 1 sun illumination was applied.

the K_2CO_3 -Tris solution compared with the pure K_2CO_3 solution, approximately four to ten times higher at 40–80 °C (Fig. 3c,d and Supplementary Figs. 10–14). For the high-concentration capture solution (Supplementary Fig. 5), -0.83 – $1.1 \text{ mol l}^{-1} \text{CO}_2$ gas could be released when heating to 60 °C using 5–15% CO_2 inlet streams (Supplementary Fig. 15 and Supplementary Fig. 16). The regeneration energy (kJ per mole of CO_2) can be calculated by $(\Delta T \times C_p \times \rho) / [\text{CO}_2]_{\text{released}}$, where ΔT is temperature increase, C_p is the specific heat of the capture solution ($-4,000 \text{ J kg}^{-1} \text{K}^{-1}$; Supplementary Fig. 17), ρ is the density of the solution (g cm^{-3}) and $[\text{CO}_2]_{\text{released}}$ is the amount of CO_2 released (mol l^{-1}). Compared with the state-of-the-art aqueous amine sorbents that require high release temperatures and energy penalty for water evaporation^{8,42} (usually ≥ 120 °C, -220 – $350 \text{ kJ mol}^{-1} \text{CO}_2$), Tris-enhanced pH swing can potentially reduce the regeneration energy consumption to -125 – 160 kJ mol^{-1} of CO_2 captured by high-concentration K_2CO_3 -Tris systems from 5–15% CO_2 sources.

A thermodynamic analysis of the closed system was carried out to further evaluate this thermally driven pH-swing feature. Starting

from the CO_2 -saturated K_2CO_3 -Tris system at a pH of 7.7, the system pH dropped to 7.1 under elevated temperatures (Fig. 3e). The concentrations of Tris and TrisH^+ returned to their initial values after CO_2 was released (Figs. 2e and 3e). Following the temperature increase, the concentration of HCO_3^- decreased and the concentration of $\text{CO}_2(\text{g})$ increased markedly, indicating the successful release of a large amount of the captured CO_2 (Fig. 3f and Supplementary Fig. 18), which aligns well with our experimental results. An evaluation of the absorption and desorption in the K_2CO_3 -Tris system indicates that ~65–85% of absorbed CO_2 can be released when the temperature reaches 60–80 °C (Supplementary Fig. 19).

Solar-powered continuous-flow CO_2 capture and release

To demonstrate the practicality of the K_2CO_3 -Tris system, we designed a continuous-flow CO_2 capture and release system powered by solar light. A tube-in-tube geometrized reactor was configured with an inner polypropylene hollow fiber to strip released CO_2 gas from the aqueous solution and an outer polyvinyl chloride (PVC) tube that was coated

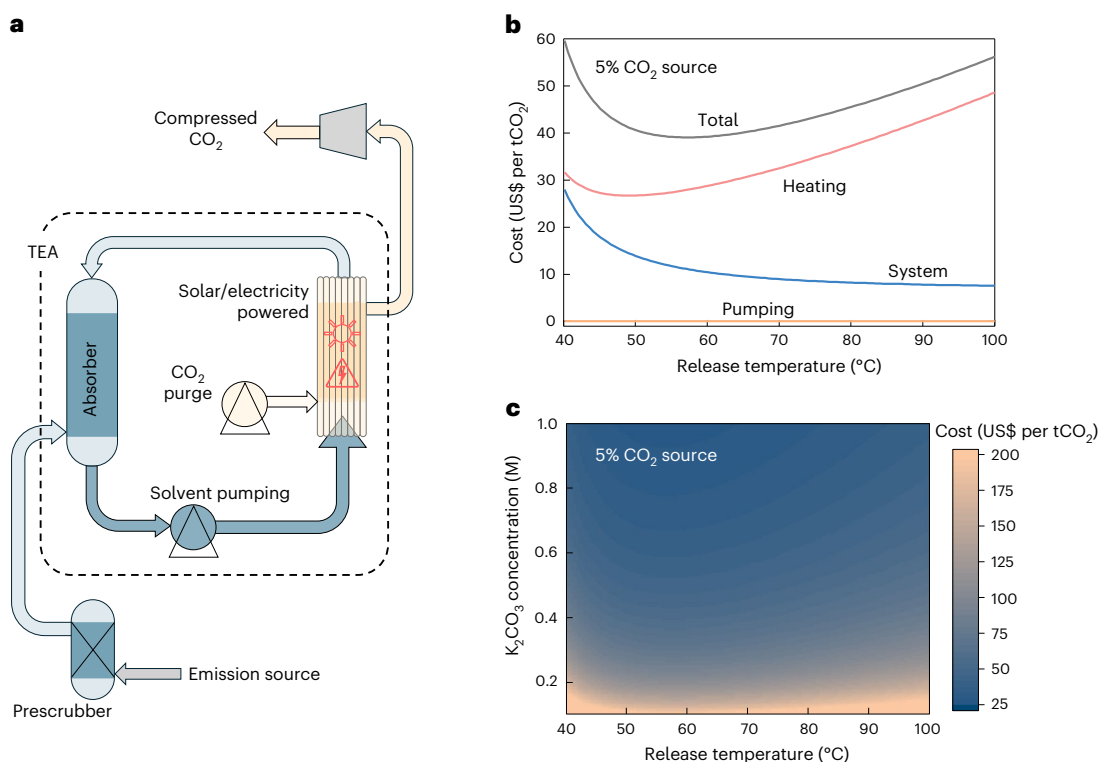


Fig. 5 | TEA of thermal pH-swing CO₂ capture and release system. a, A schematic of the proposed hybrid powered carbon capture system with solar energy and electricity; the scope of this TEA is shown in the dashed box. **b**, The components

costs including system and materials, electricity heating and pumping, at different heating temperatures. **c**, The total costs of the system under different K₂CO₃ concentrations and release temperatures using a 5% CO₂ inlet source.

with commercialized solar absorbing polymers (Fig. 4a). Then it was wound within a sealed chamber with a transparent glass front cover and an insulating ceramic back shield (Fig. 4b). The K₂CO₃-Tris solution flowed through the annular region, while a sweep stream (100% N₂ or 100% CO₂) passed through the lumen of the hollow fiber. The driving force for stripping is the difference between the dissolved and released CO₂ partial pressures, which can be optimized by flow rates of the solution and sweep stream, and the release temperature (Supplementary Fig. 20; see Methods for details). The K₂CO₃-Tris system enabled the effective regulation of CO₂ release upon 1 sun (1 kW m⁻²) irradiation, as reflected in the enhanced CO₂ concentration in the N₂ sweep stream used to emulate recovery under vacuum. The concentration increased from 1% feed to ~7% in N₂, 5% to ~20% and 15% to ~60% (Fig. 4c). Note that these concentrations in the sweep stream depend on its flow rate. Moreover, the concentration of the K₂CO₃-Tris solution can be increased to further improve the results as 1 M of K₂CO₃ with 2.5 M of Tris can concentrate 5% CO₂ feed to ~50% CO₂ using a N₂ sweep stream under 1 sun (Supplementary Fig. 21).

Considering the high cost of using solar concentrators in practical cases, we demonstrated the effectiveness of releasing CO₂ absorbed from a 1% feed by exposure to natural sunlight intensities^{43,44} (0.6–0.8 kW m⁻²) (Fig. 4d). The temperatures of the solution inside the PVC tube reached ~50–58 °C under lower solar intensity, depending on the photothermal conversion of the polymer coating and the thermal transport properties of the tubing, which can be further improved upon scaleup. The system exhibited excellent durability, with a stable concentrated CO₂ output from a low-concentration feed (1%) over 10 days (Fig. 4e).

To further verify the ability to collect pure CO₂ gas as the product, we also conducted experiments using a 100% CO₂ sweep stream (see Methods for details). Mass flow meters (MFMs) were used to monitor the mass flow of pure CO₂ at the inlet and outlet of this continuous-flow

system (Fig. 4f). The pure CO₂ stripped per volume of K₂CO₃-Tris solution was calculated from the increase in the mass flow between the two readings to be 69 and 190 mmol l⁻¹ for 5% and 15% CO₂ inlet sources, respectively (Supplementary Table 3). When the inlet CO₂ source concentration was reduced to 1%, the partial pressure of the released CO₂ in solution was insufficient to drive its desorption, and indeed, the driving force was reversed such that CO₂ was absorbed from the sweep stream, leading to a reduction in flow at the outlet. To ensure a sufficiently high solution partial pressure for effective CO₂ stripping, it was necessary to increase the solar intensity to achieve higher temperatures ≥77 °C (Supplementary Fig. 22). It may be more pragmatic to use a membraneless flash tank configuration, perhaps under a partial vacuum, to strip the released CO₂ (Supplementary Figs. 23 and 24).

TEA and further improvements

A detailed technoeconomic analysis (TEA) was conducted to evaluate the cost-effectiveness of the K₂CO₃-Tris continuous-flow carbon capture system when implemented at a large scale (Supplementary Note 2). Although a complete carbon capture system using aqueous sorbents requires the evaluation of the pre-scrubbing of emission sources and compression systems^{15,45}, this analysis focuses on the systems specific to the pH-regulator-enhanced thermal approach developed herein (Fig. 5a). It should be noted that the Tris-enabled low-temperature pH-swing process is not expected to substantially change the operating requirements of the prescrubber and subsequent compression steps; thus, analyses of auxiliary costs outside of the current system should also be considered at the commercialization stage⁴⁶.

The system was designed for an annual capture capacity of 1 MtCO₂, with the TEA examining three critical operational parameters: CO₂ release temperature (40–100 °C), sorbent concentrations (0.1–1.0 M K₂CO₃) and inlet CO₂ concentration (1–30%). The economic model incorporated both capital expenditure and operational costs,

considering key components such as membrane systems, chemical costs, pumping and heating equipment. The scaled system performance was estimated on the basis of experimental results using pure CO₂ as the sweep gas (Supplementary Table 3). The TEA considered the total cost per tCO₂ captured as determined by three components, pumping energy, heating energy and system costs, whereas the solar light is the heating source. Electricity usage was included as a backup when solar power is not available, which ensures round-the-clock operation. A capital recovery factor of 0.082 was included to annualize capital costs^{13,14}, enabling a comprehensive assessment of the system and its long-term economic viability.

The TEA results for 5% CO₂ inlet sources showed that increasing the CO₂ release temperature from 40 °C to 100 °C generally led to higher heating costs but improved overall efficiency to reduce the system costs (Fig. 5b). Their interplay resulted in an optimal release temperature at -59 °C, at which the system is the most cost-effective. It should also be noted that the pump costs in our TEA were negligible due to the high volumetric capacity of our system, which effectively reduces the amount of aqueous solution that needs to be pumped. The sorbent concentration also affects the system economics, as higher K₂CO₃ concentrations can improve CO₂ capture capacity, but probably leads to higher pumping and maintenance costs due to higher solution density and viscosity. With moderate K₂CO₃ concentrations (-0.6 M) and optimized release temperatures, the total cost per tCO₂ captured from 5% inlet sources is estimated to be -US\$39 per tCO₂ (Fig. 5c). This value can vary from -US\$187 to US\$14.5 per tCO₂ with inlet CO₂ concentration changes from 1% to 15% (Supplementary Figs. 25 and 26). Moving away from membrane-based tube-in-tube geometrized reactors toward alternatives such as flash tank and condensation column could decouple the conflicting demands of materials and manufacturing costs. Our TEA results suggested that the addition of the thermal pH regulator could greatly enhance the cost-effectiveness and overall performance of thermal-based carbon capture technologies, where further cost reductions are possible with materials and system optimization.

Conclusion

We developed an approach to address the major challenge of the energy-intensive regeneration of aqueous CO₂ absorbents by the integration of Tris as a thermally responsive pH regulator in carbonate solutions. This system exploits the controlled temperature-dependent pK_a characteristics of Tris, wherein TrisH⁺ ions serve as a strategic buffering agent to maintain basic pH conditions for enhanced CO₂ absorption at ambient temperatures. In addition, through fast-response TrisH⁺ dissociation-induced pH modulation, Tris enables effective CO₂ release at substantially reduced temperatures (≤60 °C) compared with the temperatures required for K₂CO₃-only solutions. The practical implementation of this technology through a continuous-flow reactor featuring a tube-in-tube configuration was demonstrated to be able to process diluted CO₂ streams to produce high-purity, concentrated CO₂ using only solar energy (≤1 kW m⁻²). To address the inherently slow absorption kinetics of K₂CO₃-based aqueous sorbents, larger absorption columns with optimized contactors are needed to achieve a higher degree of CO₂ removal. In summary, this system exhibited robust stability and compelling economic advantages, making it particularly suitable as a retrofit solution for existing thermal processes in industrial decarbonization applications.

Methods

Materials

Commercially available chemicals (potassium carbonate, Tris) were purchased from MilliporeSigma. All commercially available compounds were used as received. The CO₂ cylinders (1%, 5% and 15% balanced by nitrogen) were purchased from Linde Gas. The hollow fiber membranes (polypropylene Q3/2) were from 3M Capillary Membrane,

and the outer PVC tubes (Soft ND-100-65), glass sheet and ceramic sheet were purchased from McMaster.

Batch absorption and desorption experiments

In general, the flow rates and concentration of CO₂ were all controlled and/or monitored by mass flow controllers, CO₂ sensors and/or MFM. The CO₂ FT-IR sensors (ExplorIR-W 5%, W 20%, M 100% CO₂) purchased from CO2Meter and used in CO₂ capture and release tests were calibrated before each use. The MFM was used to control the bubbling rate of CO₂ into the solutions, and the CO₂ sensors were located at the outlet. The CO₂ absorption and desorption can be defined as CO_{2,outlet} - CO_{2,inlet}, where CO_{2,inlet} and CO_{2,outlet} are the amounts of CO₂ entering into the system (either the absorber or the desorber) at the inlet and exiting from the outlet, respectively. The pH was measured using a pH probe (Mettler Toledo LE422), which had been calibrated before each experiment using standard pH indicators. For the CO₂ absorption measurements, the testing chambers (4, 1.5 and 1 ml) with and without capture solution (3, 1.2 and 0.6 ml, respectively) were vacuumed (3 min) to ensure identical starting conditions for each measurement. The CO₂ gas feed (1%, 5% and 15% balanced by nitrogen) was bubbled through the solutions at a rate of 3–5 sccm (standard cubic centimeters per minute). The concentration changes in CO₂ were recorded. For batch desorption tests using the hollow fiber membrane system, the flow rate of N₂ purge gas was optimized to be 5 sccm. The CO₂-loaded feed to the tube-in-tube desorber was prepared by bubbling the CO₂ gas continuously through a reservoir at 15–20 sccm depending on the concentration of the inlet source; the liquid phase flow rate was optimized to be -0.3 ml min⁻¹ on the basis of the dimensions of the inner and outer tube-in-tube reactor with fixed transport capacity. The flow rate of N₂ purge gas was optimized to be 1.0 ml min⁻¹. The tube-in-tube reactor was located in a temperature-controlled water bath (IVYX Scientific Digital) during the desorption tests.

Solar-powered continuous-flow CO₂ capture and release experiments

The flow rates were controlled and monitored by mass flow controllers, whereas the amount of CO₂ released was inferred from CO₂ sensor (for the N₂ sweep stream), or MFM (pure CO₂ sweep) readings. Similar to batch desorption measurements, the solar-absorber coated PVC-polypropylene tube-in-tube reactor was located under the solar simulator, where the solar intensity was measured using a TES 132 Solar Power Meter (Data logging). For the absorption process, the CO₂ gas was bubbled continuously through a reservoir at 15–20 sccm depending on the concentration of the inlet source; the CO₂-loaded solution was then introduced to the tube-in-tube desorber. The N₂ sweep gas flow was optimized to be 1.0 sccm, with the corresponding liquid phase flow at 0.25 ml min⁻¹. When a 100% CO₂ sweep gas was used, the initial flow rates were set to -2.37 sccm for the 15% CO₂ inlet and -1.71 sccm for the 1% CO₂ inlet.

Data availability

The data supporting the findings of this study are available in the Article or its Supplementary Information. Additional data related to this study may be requested from the corresponding author. Source data are provided with this paper.

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Author contributions

Y.G. and T.A.H. conceived the project and designed the experiments. Y.G. performed the experimental study, thermodynamic calculations and TEA study. Y.G. wrote the article with support from T.A.H.

Competing interests

T.A.H and Y.G. are inventors on a U.S. provisional patent application (no. 63/785,467) titled, 'Thermal pH swing capture of acid gases and related systems', filed by the Massachusetts Institute of Technology. The authors declare no competing interests.

Additional information

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